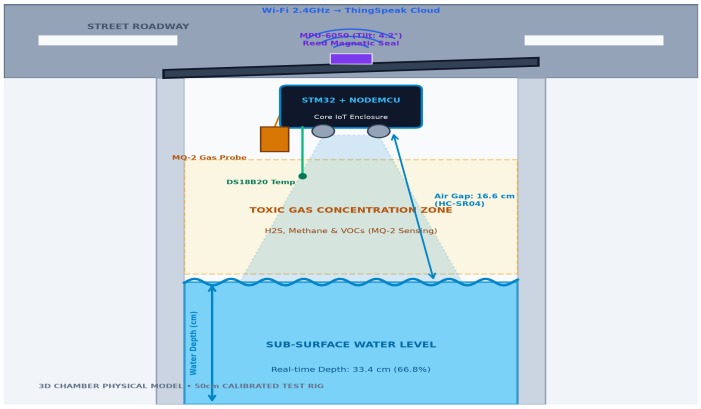
SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Infrastructure Monitoring, Real-Time Cloud Telemetry & 3D Digital Twin SCADA

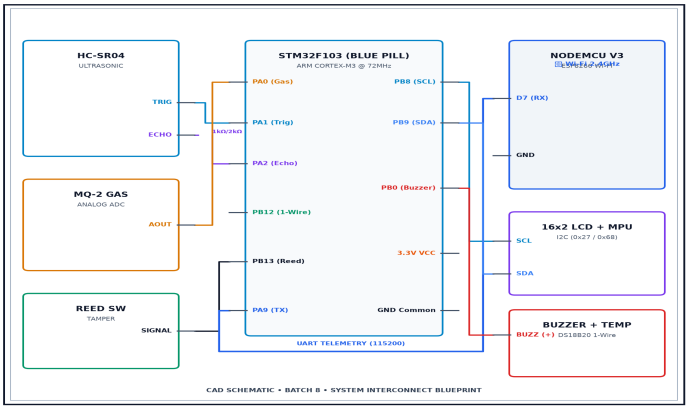
BATCH 8	2400040006 P V Kishor Kumar Reddy	2400040004 T Perumal	2400040303 Chundu Uday Sai	2400040339 Ayan Chowdhury

1. 3D SUB-SURFACE CHAMBER & SENSOR CONE



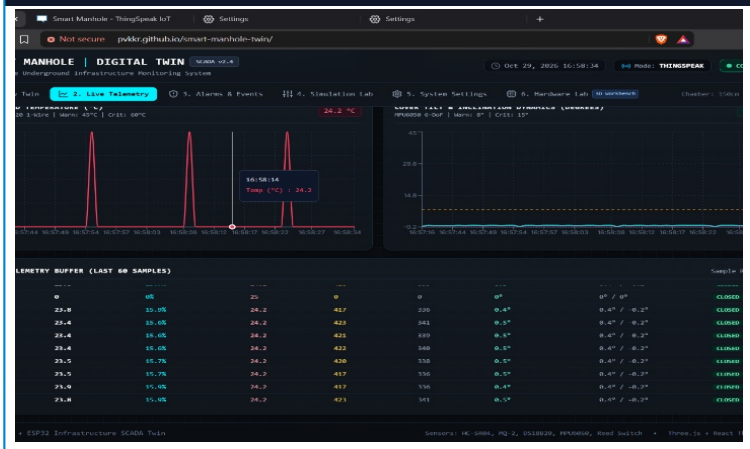
Physical 3D Model: Ultrasonic Beam, Water Level (33.4cm), Gas Zone & Enclosure

2. PROFESSIONAL CAD ELECTRONIC SCHEMATIC



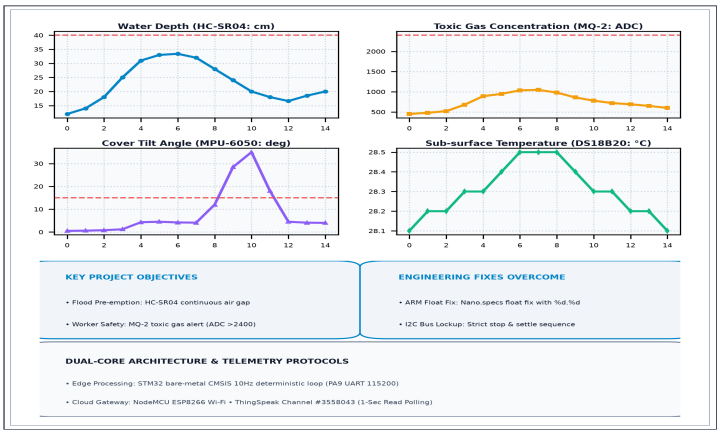
Complete Hardware Bus: Bare-Metal STM32, NodeMCU, 6 Sensors & 1kΩ/2kΩ Divider

3. 3D DIGITAL TWIN SCADA PLATFORM



WebGL SCADA Twin: Three.js physics rendering lid pitch/roll tilt & chamber surcharge

4. LIVE TELEMETRY & FIELD VERIFICATION



Multi-Sensor Analytics: Real-time curves (ThingSpeak #3558043) & telemetry specs

1. PROJECT OBJECTIVES

- **24/7 Telemetry:** Sub-surface sensing of depth, gas, tilt & temp.
- **Flood Pre-emption:** Ultrasonic surcharge tracking before street overflow.
- **Worker Safety:** Toxic gas threshold alarm (ADC >2400).
- **Anti-Theft:** 3-axis lid tilt + magnetic reed seal detection.

2. REAL-TIME APPS

- **Flood Management:** Real-time drainage surcharge alert dispatch.
- **Sanitation Safety:** Pre-entry gas screening preventing asphyxiation.
- **Road Traffic Safety:** Vehicle impact & lid displacement alarm.
- **Asset Defense:** Tamper alerts guarding fiber & power conduits.

3. BOTTLENECKS SOLVED

- **ARM Float Fix:** Nano.specs float stripped; fixed via integer %d.%d.
- **I2C Lockup:** Eliminated bus freeze with strict I2C_Stop & settle.
- **Zero-Latency:** Sub-20ms local REST + 1s ThingSpeak cache-bust.
- **5V Level Shift:** 1kΩ/2kΩ voltage divider safeguarding STM32 PA2.

4. FUTURE ROADMAP

- **NB-IoT / LoRaWAN:** Deep RF penetration through heavy iron lids.
- **Kinetic Harvesting:** Piezoelectric power from traffic vibration.
- **Edge TinyML:** Onboard neural network flow anomaly forecasting.
- **Solar Stud Tops:** Surface-mounted studs for perpetual power.